C0 — NAIVE BASELINE (evaluation reference point only)

Fixed-size chunking · dense-only retrieval · no rerank
no NPCE split · no CoVe · no numeric-verbatim guard
separate index — outside this pipeline, outside the NPCE x CoVe factorial

OFFLINE — INDEXING PLANE (runs on corpus change)

Path A Ingestion
PDF / HTML / DOCX / TXT / PPTX
OCR (if scanned) + layout/table parse

Path B Ingestion
XBRL / CSV / XLSX / DB export
schema map + row templating

Proposition Extraction
Call Site 1 — Segmenter (LLM)
Call Site 2 — Numeric/Unit/Period Extractor (bounded LLM, digits copied verbatim)

Metadata Tagging + Index Writers
deterministic + embedding model
builds the three-index invariant

Shared Indexes & Registry
Dense Index · Sparse Index (BM25)
Metadata Store (Postgres) · Placeholder Registry

User Query

serves

serves

serves

ONLINE — QUERY PLANE (runs per question)

Query Analysis + NPCE Split
collision detection (lexicon)
Call Site 3 — Query Splitter (LLM slots)
+ placeholder remap (DET)

Retrieval Core
metadata filter (DET) + hybrid retrieve (BM25+embedding) + fusion (DET) + cross-encoder rerank (EMB)

Fact Curation
Call Site 4 — FFF Rewriter (LLM)
+ numeric-verbatim guard (DET)
+ Fact-set/Query Pairer (DET)

Generation
Call Site 5 — Generator (LLM)
+ Draft Assembler (DET) → draft D

Chain-of-Verification (CoVe)
Call Site 6 — claim extraction + independent re-check (LLM, draft excluded) + deterministic numeric comparison (DET, never an LLM judgment)

Output Assembly
Abstention Gate + Placeholder Reverse-Mapper + Citation Binder
— all deterministic

Answer + Verification Trace
+ Citations (or Abstention)

LEGEND

Deterministic block
no model involved

Contains an LLM call site
(numbered, generative)

Mixed block:
deterministic + embedding / non-generative model

Shared index / registry

External reference,
not part of the pipeline

Financial RAG — Noun-Phrase Collision Elimination + Chain-of-Verification
System Block Diagram